

there is bound to be a survivor, and guess what: if you are here talking about it, you are likely to be that particular one (notice the “condition”: you survived to talk about it). So we can no longer naïvely compute odds without considering that the condition that we are in existence imposes restrictions on the process that led us here.

Assume that history delivers either “bleak” (i.e., unfavorable) or “rosy” (i.e., favorable) scenarios. The bleak scenarios lead to extinction. Clearly, if I am now writing these lines, it is certainly because history delivered a “rosy” scenario, one that allowed me to be here, a historical route in which my forebears avoided massacre by the many invaders who roamed the Levant. Add to that beneficial scenarios free of meteorite collisions, nuclear war, and other large-scale terminal epidemics. But I do not have to look at humanity as a whole. Whenever I probe into my own biography I am alarmed at how tenuous my life has been so far. Once when I returned to Lebanon during the war, at the age of eighteen, I felt episodes of extraordinary fatigue and cold chills in spite of the summer heat. It was typhoid fever. Had it not been for the discovery of antibiotics, only a few decades earlier, I would not be here today. I was also later “cured” of another severe disease that would have left me for dead, thanks to a treatment that depends on another recent medical technology. As a human being alive here in the age of the Internet, capable of writing and reaching an audience, I have also benefited from society’s luck and the remarkable absence of recent large-scale war. In addition, I am the result of the rise of the human race, itself an accidental event.

My being here is a consequential low-probability occurrence, and I tend to forget it.

Let us return to the touted recipes for becoming a millionaire in ten steps. A successful person will try to convince you that his achievements could not possibly be accidental, just as a gambler who wins at roulette seven times in a row will explain to you that the odds against such a streak are one in several million, so you either have to believe some transcendental intervention is in play or accept his skills and insight in picking the winning numbers. But if you take into account the quantity of gamblers out there, and the number of gambling sessions (several million episodes in total), then it becomes obvious that such strokes of luck are bound to happen. And if you are talking about them, they have happened to you.

The *reference point argument* is as follows: do not compute odds from the vantage point of the winning gambler (or the lucky Casanova, or the endlessly bouncing back New York City, or the invincible Carthage), but from all those who started in the cohort. Consider once again the example of the gambler. If you look at the population of beginning gamblers taken as a whole, you can be close to certain that one of them (but you do not know in advance which